

Saksham Mittal

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EDUCATION

Netaji Subhas University of Technology (Formerly NSIT), Delhi

Aug 2022 – May 2026

Bachelor of Technology in Information Technology, Minor in Artificial Intelligence

CGPA: 9.02/10

- **Coursework:** Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object Oriented Programming, Artificial Intelligence, Machine Learning, Deep & Reinforcement Learning

WORK EXPERIENCE

Writesonic (YC S21) | *Software Engineer Intern*

Jan 2026 – Jun 2026 | San Francisco, CA

- Owned the **GEO Ads** feature end-to-end, building backend services, third-party ad API integrations, and the **React/Next.js** UI, and shipping to **production in 4 weeks**; the feature shows brands their ad share-of-placement and placement rank across LLM surfaces (ChatGPT, Google AI Overviews) versus competitors.
- Added a **Top Themes** preview to the **Sentiment Analytics** platform, surfacing the key topics flagged for a brand from the main themes table for at-a-glance competitive insight.
- Built **React/TypeScript** interfaces for the **Keyword Ranking Distribution** tool, an SEO feature that shows brands where their site ranks on the SERP for a target keyword.
- Designed and shipped a **password-policy enforcement system** end-to-end, covering technical design doc, backend validation, and **Datadog monitoring**, and implementing **4 account-security controls**: password-complexity rules, failed-attempt lockout with forced reset, last-10 password-reuse prevention, and a corrected reset-email flow.
- Performed **root-cause analysis** on production API failures and cross-browser client-side errors using **Chrome DevTools**, restoring functionality across affected workflows.

Ericsson | *Software Engineer Intern*

July 2025 – Aug 2025 | Noida, India

- Built and configured automated **CI/CD pipelines** in **GitLab**, automating build, test, and deployment steps to streamline development workflows.
- Used **Docker** for containerization and **Ansible** for configuration management and deployment orchestration, supporting reproducible, reliable releases.

PROJECTS

Collab Task Board | [GitHub](#)

June 2026

- Built a real-time collaborative task board (Trello/Linear-style) with a layered **FastAPI** backend, **React/TypeScript** frontend, and **PostgreSQL** via **SQLAlchemy/Alembic**.
- Implemented stateless **JWT auth**, board-level access control, and **multi-user sync** over **WebSockets**, so concurrent users see changes live without refreshing.
- Designed a fractional-positioning algorithm for drag-and-drop reordering (**O(1)** writes per move); added **47 automated tests** (pytest + vitest) in a **GitHub Actions CI** pipeline.

Credit Card Fraud Detection System | [GitHub](#)

June 2025

- Built a fraud analytics platform processing **500,000+ transactions** using **Python, PostgreSQL, and Docker**, with ETL pipelines that run automated validation, schema-consistency checks, and data-quality monitoring; researched a production scale-out path on **Hadoop, Hive, AWS, and Spark**.
- Trained **Random Forest** and **Logistic Regression** models with **SMOTE** class balancing, achieving **AUC > 0.99** on holdout data for fraud detection and risk scoring.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, SQL, HTML, CSS

Software Engineering: Object-Oriented Design (OOD), Design Patterns, SDLC, Agile Development, Microservices, Distributed Systems, REST APIs, WebSockets

Frameworks: React, Next.js, FastAPI, SQLAlchemy, Pydantic, Bootstrap

Cloud & DevOps: AWS, Docker, Git, GitHub, GitLab, CI/CD, GitHub Actions

Databases: MySQL, PostgreSQL, MongoDB, Alembic, DBever

Developer Tools: Claude Code, Cursor, Postman, Chrome DevTools, Datadog, Playwright, Pytest, Vitest

PUBLICATIONS

ICT4SD 2026 (Springer LNNS) - "A Lightweight Two-Stage Authentication Framework for MQTT Using Cuckoo Filter Gating and One-Time Key Binding": Strengthened IoT **data security** and **privacy** by **mitigating 8+ attack vectors**, including **unauthorized access**, **replay attacks**, and **session hijacking**, with minimal resource overhead.

ACHIEVEMENTS AND EXTRACURRICULAR ACTIVITIES

- **Department Rank 1** in **Semester 3,4,7 and 8**. Received the **Talent Incentive Scheme Scholarship** for securing department Rank 1 in sophomore and senior year.
- **Activities:** IEEE NSUT, CodeChef NSUT, TechWeek'23 by IEEE NSUT, Innohacks Hackathon'24